

| Dissertation Proposal

Title: Integrating Neural Physics Engines into Embodied Foundation Models for Enhanced Physical Reasoning and Robust Generalization in Embodied Agents

Principal Goal: The principal goal of this research is to resolve the generalization bottleneck in embodied agents navigating complex, unseen physical environments by embedding differentiable Neural Physics Engines into the decision-making pipelines of Embodied Foundation Models (EFMs). This aims to endow agents with physical intuition, enabling zero-shot generalization in contact-rich and deformable object manipulation tasks, thereby bridging the sim-to-real gap.

Background & Problem Statement:

While current probabilistic Large Language Models (LLMs) demonstrate exceptional proficiency in semantic analysis and contextual language understanding, they struggle to serve directly as the "brains" of AI robotics. Existing models primarily rely on statistical probabilities for text or action prediction and lack a fundamental understanding of physical laws. Consequently, when an agent faces complex, dynamic, and unseen physical environments, its generalization capability degrades severely.

This is particularly evident in tasks involving deformable objects (e.g., folding garments) or high-precision force control (e.g., cable insertion). In these scenarios, traditional gradient-free reinforcement learning is highly inefficient and prone to execution failure due to predictive deviations. Therefore, successfully injecting deterministic physical laws into probabilistic neural foundation models is a critical challenge for the next decade of embodied AI.

Proposed Research & Methodology:

This project proposes a universal paradigm combining computational linguistics and physical simulation. Key research components include:

- 1. Integration of Neural Physics Engines:** Diverging from traditional black-box exploration, this project investigates utilizing differentiable physics simulators. By doing so, physical gradients (e.g., gravity, friction, deformation dynamics) can be directly backpropagated to the foundation model, significantly accelerating learning efficiency and physical interpretability.

2. **Physical Chain-of-Thought (CoT):** Inspired by the reasoning patterns of models like DeepSeek-R1, we will design a "Physical CoT" mechanism. Before outputting an action token, the model will be forced to sequentially analyze physical properties (e.g., *Observation: fragile material -> Constraint: torque limit -> Action: motion planning*), enhancing the interpretability of decision-making.
3. **Spatial RAG & Episodic Memory:** Traditional text-based Retrieval-Augmented Generation (RAG) will be transformed into 3D Scene Graphs. By leveraging vector databases and computing element clustering, the robot will perform real-time reasoning based on spatial geometry and historical physical interactions rather than just linguistic context.
4. **Target Scenario Validation:** The evaluation will focus on contact-rich and deformable object manipulation. We will validate the architecture's effectiveness in zero-shot or few-shot execution of highly complex tasks, such as garment folding or socket insertion.

Expected Outcomes:

1. A systematic framework for injecting physical rules into the reasoning pipelines of Embodied Foundation Models.
2. Development of a simulation evaluation environment (integrating frameworks like Unity or Three.js) supporting physical gradient backpropagation and reinforcement learning.
3. Experimental validation demonstrating that this architecture achieves higher generalization success rates in handling deformable objects compared to pure data-driven models.

Completion Criteria:

- A baseline completion involves a thorough workload characterisation and reproduction of existing benchmarks. This will include characterizing existing embodied AI baselines and reproducing the Physical CoT's improvement on simple grasping tasks within a simulated environment.
- An outstanding completion involves a novel systems contribution —namely, a physics-engine-driven Embodied Foundation Model. This includes rigorous experimental evaluation demonstrating superior zero-shot generalization capabilities in complex

deformable object manipulation tasks, alongside preliminary sim-to-real transfer validation.

Citations

- @inproceedings{10.1145/3731569.3764843, title = {KTransformers: Unleashing the Full Potential of CPU/GPU Hybrid Inference for MoE Models}, author = {Chen, Hongtao and Xie, Weiyu and Zhang, Boxin and Tang, Jingqi and Wang, Jiahao and Dong, Jianwei and Chen, Shaoyuan and Yuan, Ziwei and Lin, Chen and Qiu, Chengyu and Zhu, Yuening and Ou, Qingliang and Liao, Jiaqi and Chen, Xianglin and Ai, Zhiyuan and Wu, Yongwei and Zhang, Mingxing}, booktitle = {Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles}, year = {2025} }
- *Additional citations focusing on Differentiable Physics (e.g., Brax, DiffTaichi), spatial RAG implementations, and embodied foundation models to be added upon comprehensive literature review.*

Additional Information

Category	Details
Difficulty	3 - Hard
Desirable Skills	CUDA programming, familiarity with hardware. Experience with differentiable physics simulators (e.g., MuJoCo, Brax), Unity environment development, deep understanding of underlying mechanisms for LLMs (e.g., LangChain, vector databases).
Essential Skills	Python and/or C++ programming, understanding of computer architecture and memory hierarchy, familiarity with PyTorch and LLM architectures, basic knowledge of profiling and performance analysis. Familiarity with multimodal/embodied foundation model architectures.
Resources Required	GPUs, potentially CPUs for simulation. Computation clusters for differentiable physics simulation, potentially physical robotic arms for real-world (sim-to-real) evaluation.